

# ■ ORCA Marine Intelligence Platform

Simple Layman's Project Guide & Process Breakdown (Q&A; Format) | ISRO SIH26176

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## Q1: What is this project all about in plain words?

Think of ORCA as a smart **AI Ocean Assistant** designed for fishermen, boat captains, coastal guards, and marine researchers in India.

Just like asking Google Maps for directions to a restaurant, anyone can ask ORCA questions about the sea in simple everyday words—such as *'Where can I catch the most fish today near Kochi?'* or *'Is the sea safe near Mumbai right now?'*—and get instant, friendly advice along with a live interactive map.

## Q2: What main real-world problems does ORCA solve?

ORCA addresses four major challenges faced by coastal communities:

- **Finding Fish Easily:** Saves fishermen time and expensive boat fuel by highlighting exact ocean areas (Potential Fishing Zones) where fish are gathering.
- **Keeping Boaters Safe:** Warns boat captains about dangerous high waves, strong winds, or sudden storm warnings.
- **Preventing International Border Crossings:** Alerts fishermen if they drift too close to international sea boundaries (like Sri Lanka, Pakistan, or Bangladesh) to avoid arrests or fines.
- **Removing Language Barriers:** Operates in 10 local Indian coastal languages so fishermen can speak or listen in their mother tongue.

## Q3: What are the main parts of the system that were built?

The system consists of 4 main building blocks:

1. **The Visual Map (Frontend):** An interactive map displaying colored shapes—green for fishing zones, red for storm warnings, purple for sea boundaries, and cyan for safe boat routes.
2. **The AI Brain (Backend Agents):** Smart digital assistants that calculate sea temperatures, weather risks, and safe navigation paths.
3. **The Multi-Lingual Translator:** Converts questions typed or spoken in regional Indian languages into English for the AI, and translates the answer back.
4. **The Natural Human Voice Engine:** Speaks the answers out loud in realistic, human voice accents (Tamil, Hindi, Marathi, Bengali, etc.) so boaters can just listen.

## Q4: How does it work step-by-step when a user asks a question?

Here is the exact journey of a question:

- **Step 1 (Ask):** The user types or speaks a question (e.g. *'Show me safe routes near Goa'*).
- **Step 2 (Translate):** The system detects the language (e.g. Konkani or Marathi) and converts it for internal processing.
- **Step 3 (AI Analysis):** Smart agents analyze sea coordinates, wind speeds, wave heights, and fish concentration data.
- **Step 4 (Map Fly & Draw):** The map automatically zooms directly to Goa and draws colored shapes showing safe path lines.

- **Step 5 (Speak Back):** The system speaks the response out loud in a natural human regional voice.

### Q5: Which languages and coastal regions are covered?

- **11 Supported Languages:** Hindi, Gujarati, Marathi, Konkani, Kannada, Malayalam, Tamil, Telugu, Odia, Bengali, and English.
- **All 7,516 km of Indian Coastline:** Covers 35+ major ports and harbors across Gujarat, Maharashtra, Goa, Karnataka, Kerala, Tamil Nadu, Andhra Pradesh, Odisha, West Bengal, Lakshadweep, and Andaman & Nicobar.

### Q6: How was the app built and deployed on Vercel?

- **Frontend:** Built using React & MapLibre GL for smooth map rendering.
- **Backend:** Built with FastAPI (Python) & LangGraph for AI multi-agent orchestration.
- **Deployment:** Configured for instant deployment on Vercel with automatic mock data fallback (so it runs cleanly without needing complex satellite API setup).

**Project Summary Status: Fully Built & Ready for Vercel Demo | ISRO SIH26176**